Diego Villagran Salazar

ETL Developer | Data Engineering | Pipeline Automation
diegovillasal@gmail.com | +52 55 7207 1183 | www.dvillagrans.dev | LinkedIn | GitHub

Professional Summary

Data Science student with hands-on experience building ETL pipelines, orchestration workflows, and automated data integration systems in production environments. Skilled in SQL optimization, Python-based data processing, and containerized data infrastructure handling 300+ daily inputs and 10K+ service requests. Strong foundation in data modeling, batch processing, and monitoring for reliability at scale. Currently expanding into enterprise ETL tools such as Ab Initio.

Education

Bachelor of Science in Data Science

Superior School of Computer Science (ESCOM-IPN) *2022 – 2026 (December)*
Mexico City, Mexico

- Maintained GPA of 8.5/10.0 while completing 45+ credit hours in advanced coursework
- Specialized in Data Engineering, ETL Pipelines, Database Systems, Big Data Analytics, and Statistical Modeling

Professional Experience

AI & Automation Intern

EyeNet *April 2025 – Present*
Remote

- Designed and maintained ETL pipelines using Python and orchestration workflows to process 300+ documents daily with 92% extraction accuracy, transforming unstructured data into PostgreSQL databases
- Built data integration workflows connecting OpenAI and Gemini APIs with database systems, automating extraction and transformation of 200+ documents per day
- Developed containerized data services with PostgreSQL, Redis, and MongoDB, supporting 5+ services handling 10K+ daily requests with optimized query performance
- Implemented CI/CD pipelines and monitoring systems with automated alerts, reducing deployment time by 50% and catching 95%+ of data quality issues before production
- Created SQL optimization strategies and database indexing improvements, reducing query response times by 40% (850ms to 510ms)

Projects & Research

Qalma - Data Processing Platform for Wellness Analytics

Highlighted Project *2025*

- Built real-time data ingestion pipeline processing 256 Hz EEG signal streams from 4 channels with <100ms latency, serving 50+ users
- Developed ETL workflows using Python-based data services (Flask/FastAPI) to extract, transform, and load sensor data into PostgreSQL, handling 1K+ daily data points per user
- Implemented batch processing jobs for historical data analysis and aggregation, enabling predictive analytics with 85%+ accuracy
- Designed cloud data infrastructure with Supabase and PostgreSQL achieving 99.9% uptime, with automated backup strategies using 3-2-1 methodology
- Created data quality monitoring dashboards tracking pipeline health, data freshness, and processing metrics across 100+ active sessions

Microservices Data Orchestration & Monitoring

DevOps & Data Engineering Project *2025*

- Orchestrated Docker Compose environment with 8+ data services (PostgreSQL, Redis, Nginx, APIs) achieving 99.5% uptime with automated health checks
- Built automated backup and recovery system using pg_dump and rsync with 3-2-1 strategy, reducing recovery time from 2 hours to 15 minutes
- Designed monitoring infrastructure with Grafana and Prometheus tracking 25+ data pipeline metrics, detecting 95%+ of issues before production impact
- Performed load testing and optimization using Apache Bench, improving data throughput and reducing API response time by 40%
- Implemented log aggregation and analysis system for tracking data flow, errors, and performance bottlenecks across distributed services

Technical Skills

ETL & Data Engineering: ETL Pipelines, Data Integration, Batch Processing, Data Warehousing, Star Schema/Fact-Dimension Modeling, Data Marts
Databases: PostgreSQL, MongoDB, Redis, Supabase, Database Indexing & Query Optimization
Programming & Automation: Python, SQL, Bash, REST APIs, Orchestration Workflows (n8n, CI/CD)
Cloud & DevOps: Docker, Microservices Architecture, Git, AWS/Azure, Linux Administration
Monitoring & Reliability: Grafana, Prometheus, Log Aggregation, Data Quality Monitoring
ML & Analytics: Pandas, NumPy, Scikit-learn, TensorFlow, Feature Engineering, Power BI, Tableau

Certifications & Languages

Certifications: Build a Data Warehouse with BigQuery, Engineer Data for Predictive Modeling with BigQuery ML, Manage Kubernetes in Google Cloud, Google AI Essentials (Coursera)
Languages: Spanish (Native proficiency), English (B2 in-course)